

Systems that refuse to invent.

Kunal Kumar

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MATS round 2 · Activate AI Fellow – India's first cohort, 1 of 15 from 500+
HPAIR 2026 delegate, Harvard

Every system I use will guess, then deliver the guess in the voice it uses for facts. That isn't a bug – **“I don't know” is harder to build than a confident answer**, and nobody is rewarded for building it. So that is what I build instead, and I have now done it enough times to admit it is a thesis rather than a habit.

What follows is not five apps. It is **five parts of one person. 32 repositories · 829 contributions in twelve months · 29 patches merged into 35 open-source projects.**

01 AI Safety MATS · Activate Fellow ATTEST · Triage · Watershed	02 Recall Local-first continuity OS shipped as a .dmg	03 Upstream 29 merged patches 10 maintainers · 4 languages	04 Community 500+ participants 50+ students taught	05 Kunal HQ Personal life OS a score that can't be cheated
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Kunal Kumar	Computer Science · Student Class of 2028	kunalsain0324@gmail.com · +91 85448 88032 linkedin.com/in/sainkunal · github.com/kunalKumar-13	Apple Developer Academy Indonesia @ BINUS, Bali · Cohort 2027 · Tech track

YEAR ACCOMPLISHED

2026 — ongoing

ROLE / POSITION

Sole author of all three systems.
Self-initiated.

PUBLICATION LINK

github.com/kunalKumar-13/attest

github.com/kunalKumar-13/deterministic-support-triage

One question sits under everything I build: **when a capable model proposes something, what is allowed to believe it?** What decides whether a system is safe to deploy is structural: what the model is architecturally *permitted to cause*.

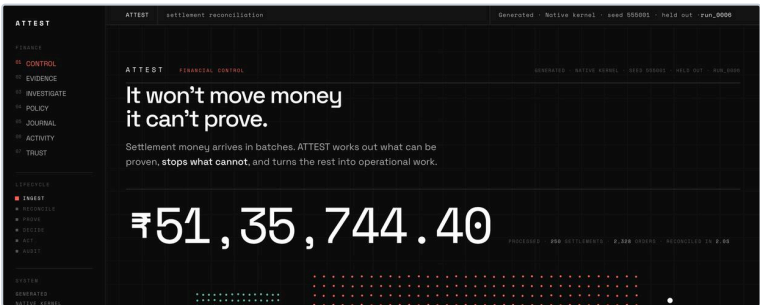
I'm through to **round 2 at MATS** and was selected as an **Activate AI Fellow** — India's first cohort, 1 of 15 from 500+. Three systems I had already built turn out to be about the same problem.

Technical aspects

ATTEST — a payment gateway sends one lump sum covering many orders, and more than one set of orders can add up to the same credit *exactly*. Those sets discharge different customers, so guessing marks the wrong customer as paid while the books still balance. The advisor proposes which records to look at and is **never given an amount**; a verifier that **shares no code with it** re-derives every explanation; the ledger records nothing unproven. Result: **82.4% correctly refused, ₹99.5 lakh deliberately held, ₹0.00 wrongly posted** across 500 held-out settlements. Python with a Rust solver kernel.

Support Triage — code owns the control flow. The LLM is invoked **once per ticket inside an unforgeable sentinel block** and returns typed JSON a policy validator adjudicates; it cannot decide escalation, execute tools, or see raw PII. Defends direct and multilingual prompt injection (7 languages) plus zero-width, homoglyph and base64 obfuscation. Worst case is *escalate to a human*, never a destructive action. **132/132 tests; two runs byte-identical, verified by SHA-256**.

Watershed — a multi-agent RL arena, GRPO over **100k+ self-play episodes** with trust dynamics and coalition formation in the reward. Run it long enough and the agents stopped competing: **alliances formed**. Nobody put them there.



HELD-OUT EVALUATION · SEED 555001

Proven	39 of 250 — eligible to automate
Held	210 — ambiguous, refused
Contradicted	1
Wrongly posted	₹0.00
Baseline	benchmark names a matcher more precise than mine

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YEAR ACCOMPLISHED
2026

ROLE / POSITION
Sole author — engine,
launcher,
browser extension, daemon,
docs.
Self-initiated.

PUBLICATION LINK
web-seven-puce-99.vercel.app

github.com/kunalKumar-
13/Recall-me

docs.recall.computer

Someone interrupts you at 3pm. You sit back down and everything is still there — the file, the tab, the branch — and none of it helps. **You didn't lose files. You lost continuity.**

Spotlight and grep assume you remember a string. A cognitive restart never asks that. It asks *what was I doing, which conversation was that in, where did I leave the implementation* — and nothing on the machine answers those.

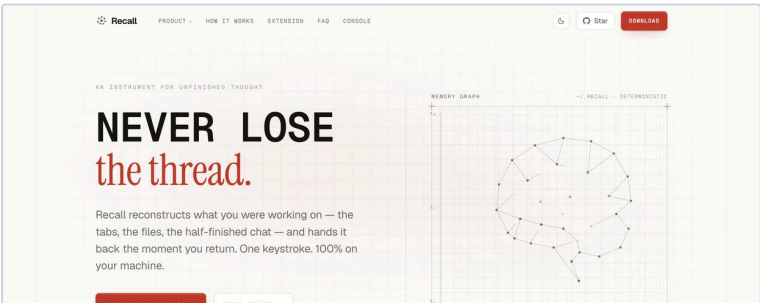
Technical aspects

Seven composed layers over a local event log: **events** → **sessions (30-min gaps)** → **micro-contexts** → **resurfacing** → **threads** → **evolution** → **recovery**. A 200-line Chrome extension posts activity to a local API at 127.0.0.1:4545; watchdog reindexes a saved file within ten seconds; semantic file search runs all-MiniLM-L6-v2 over ChromaDB.

One keystroke returns the moment, the session *and* the micro-context from one half-remembered phrase. An empty query gives a *Continue where you left off* card, **capped at four items, no badges, no feed**.

The decision that shaped it: no LLM in the production path. Determinism, so the same question can't return two different pasts — enforced by a 35-section smoke test. And privacy you can check rather than be promised: no cloud, no account, no telemetry. Everything it knows lives in `~/recall/` as plain JSON, readable with `cat`, deletable with `rm`.

Ships two ways: from source (PyQt6 + FastAPI) and as a **~26 MB Tauri menu-bar app** that honestly reports semantic search as *disabled* rather than silently returning worse results. **1,003 files**.



DESIGN CONSTRAINTS, CHOSEN UP FRONT

Deterministic	same events in → same output out
Local only	no cloud, account or telemetry
Install	three minutes, five commands
Degradation	announced, never silent

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YEAR ACCOMPLISHED

2025 – 2026

ROLE / POSITION

Outside contributor.
Self-initiated, entirely in public.

PUBLICATION LINK

github.com/pulls?q=is:pr+author:kunalKumar-13+-user:kunalKumar-13

Every other entry here is work I graded myself. Upstream, “done” is when someone who has maintained a codebase for a decade – who did not ask for my patch and owes me nothing – decides it is good enough to carry.

29 merged across 35 repositories. 12 approvals from 10 maintainers in 7 organisations. It is also the closest thing I have done to a ten-month cohort: sustained collaboration with people who don't work for you, under conventions you must learn before you're entitled to an opinion.

Technical aspects

Deliberately spread across ecosystems that share nothing – a children's music environment, a Bitcoin Lightning wallet, the OpenStreetMap editor, Django's own website, scientific data QC, and the **Fortran standard library**. Four languages, six review cultures.

Almost none are features. `musicblocks#8495` releases a busy state so a widget cannot deadlock. `#8589` raises `NanError` in every operation that promises it. `plone.versioncheck#79` parses buildout files with `zc.buildout`'s own parser instead of by hand. `id#12794` detects `goo.gl` shortlinks as an incompatible source, a licensing trap for mapmakers. **This is maintenance work – what decides whether software still runs in five years.**

Fortran is the one I'd point at: I had no reason to know it, and landing a patch meant learning how numerical scientists argue about correctness before writing a line. **The skill was arriving somewhere I had no standing and becoming useful anyway.**

WHERE THE WORK LANDED · VERIFIED AGAINST THE GITHUB API, 11 SEP 2026

8 · sugarlabs/musicblocks	JavaScript – a music-and-programming microworld for children
3 · django/djangoproject.com	Python – the site Django itself runs on
3 · metabrainz/listenbrainz	Python + TypeScript – open music listening data
2 · plone/volto · 2 · ZeusLN/zeus	JavaScript · TypeScript – a CMS frontend, a Bitcoin Lightning wallet
1 · fortran-lang/stdlib	Fortran – the Fortran standard library
+ 5 more	openstreetmap/id, plone/meta, plone.versioncheck, fortran-lang/webpage
Maintainers	walterbender · MonkeyDo · sarahboyce (Django Fellow) · kaloudis · tyrasd · SaptakS · jensens · gforcada · jalvesz · mauritsvanrees

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YEAR ACCOMPLISHED

2024 – 2026

ROLE / POSITION

Growth & Ops Lead · Teaching Assistant · Operations Lead · Head of Immersions.
Roles held, not a built artefact.

PUBLICATION LINK

linkedin.com/in/sainkunal

A ten-month cohort is not a course. It is twenty-odd people in a room for a year, and whether that room produces anything depends on things that aren't engineering – whether the logistics hold, whether people who think differently can hear each other, whether someone notices the person who has gone quiet.

Somebody has to be that person. I have been, for two years.

Technical aspects

Hack Club, 16 months. Regional technical infrastructure and operational logistics for Campfire, Daydream and CodeDay Bengaluru – **500+ participants fed and online.** At that scale every failure is one you didn't plan for; the wifi never goes down the way you tested. So you stop trying to prevent failure and start designing the degradation – what still works, and who finds out first. **Every system in this portfolio has a documented failure mode, and that habit came from a wifi outage, not a textbook.**

Teaching, 50+ students. The first explanation is the one you already have; the second is the one you prepared. The third is the one you build live, out of whatever the student just told you they *do* understand – and it is the only one that works. It changed how I write: every README I ship opens by naming what the software doesn't do, because people can only hold a new idea once they know its edges.

Now: Operations Lead of the Open Source Club and Head of Immersions for Aspirants Relations – immersions being the closest thing I've run to what the Academy does. **HPAIR 2026 at Harvard** put me in rooms arguing about things I had no technical claim on.

ROLES

Nov 2024 – Mar 2026	Growth & Ops Lead, Hack Club – Campfire, Daydream, CodeDay Bengaluru · 500+ participants
May – Jun 2026	Teaching Assistant, Data Structures & Algorithms – 50+ students mentored
2026 – present	Operations Lead, Open Source Club
2026 – present	Head of Immersions, Aspirants Relations
2026	Selected delegate, HPAIR – Harvard Project for Asian and International Relations

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YEAR ACCOMPLISHED

2026 — in daily use

ROLE / POSITION

Sole author — vault, capture command, sorting agent, dashboard, game layer.
Self-initiated. Private repository.

PUBLICATION LINK

Local demo · private repo

github.com/kunalKumar-13

July 2026: a full-time internship, a degree, three applications in flight, and a life. All of it held in working memory, because there was nowhere else to put it.

It wasn't laziness — **120 LeetCode problems solved, GitHub days with 24 contributions**. The result was **fog**: too many open loops with no place to land.

Technical aspects

The diagnosis everyone offers is discipline. It doesn't work, for a specific reason: **every capture tool charges its tax at the exact moment you have nothing to give**. The thought arrives mid-task; capturing it means choosing a project, a tag, a type — and *deciding* is precisely the resource you've run out of.

So capture is one line from any terminal — `brain <thought>` — appended to a queue, returning immediately. **Two seconds, zero judgment, sorted later**. Three organs: a **vault** of plain markdown (the source of truth), an **agent** that triages it, and a **page** showing only what matters now.

A macOS LaunchAgent fires at **8:23 every morning**, drains the queue and pulls the world — weather, both GitHub accounts, LeetCode, calendar, reminders, Mac vitals — then settles XP and rebuilds the dashboard. **Every source fails independently** — a dead VPN never blocks the weather.

The rule that makes it work: XP is earned only from real, verified data, never self-reported. A number that is cheap to move is a number you will stop believing. I cannot lie to it — which is the only reason I still do.

THE SYSTEM

Vault	~/LifeOS — plain markdown. Archive, never delete. Absolute dates only.
Capture	<code>brain <thought></code> — two seconds, no decisions
Pulse	<code>launchd</code> , 08:23 daily, headless, sources fail independently
Surfaces	Today · Work · Life · Game · Manifest — default-collapsed is law
Status	v10 · in daily use · runs my mornings

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